Google Play Store

Exploratory Data Analysis & SQL

Business Challenge: EDA & SQL · 10 Business Questions
Dataset: Google Play Store (Kaggle) · 2,312,944 applications

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1. Dataset Introduction

The Google Play Store dataset was sourced from Kaggle and contains information covering **2,312,944 Android applications**. Each record captures a comprehensive snapshot of an app's metadata at scrape time, covering commercial, technical, and audience-related attributes. The dataset is notably clean — no missing values — making it ideal for direct SQL analysis and exploratory visualisation.

Dataset Attributes — 24 Columns

Attribute	Description
App Name	Application display name
App Id	Unique package identifier (e.g. com.example.app)
Category	Store category (e.g. Games, Tools, Finance)
Rating	Average user rating (1.0 – 5.0)
Rating Count	Total number of user ratings submitted
Installs	Install range string (e.g. '1,000,000+')
Minimum Installs	Lower bound of install count (numeric)
Maximum Installs	Actual install count (numeric)
Free	Boolean — app is free to download
Price	Purchase price (0.0 for free apps)
Currency	Price currency code (e.g. USD)
Size	APK size (e.g. '12M', 'Varies with device')
Minimum Android	Minimum required Android OS version
Developer Id	Developer's unique store identifier
Developer Website	URL to the developer's public website
Developer Email	Developer's contact e-mail address
Released	Original app release date
Privacy Policy	URL to the app's privacy policy page
Last Updated	Date of the most recent app update
Content Rating	Audience age group (Everyone / Teen / Mature 17+)
Ad Supported	Boolean — app contains advertising
In App Purchases	Boolean — app offers in-app purchases
Editor's Choice	Boolean — featured as a Google Editor's Choice

Attribute	Description
Scraped Time	Timestamp of when the app record was scraped

2. Data Cleaning & Assumptions

Before running any SQL queries, the dataset was loaded into a pandas DataFrame for validation and light preprocessing. The following steps were applied:

No Nulls or Missing Values

The dataset contained no null or missing values — it is exceptionally clean. No rows needed to be dropped for completeness issues.

Dropped Broken System App Rows

Some system apps (e.g. Google Camera) showed Maximum Installs = 1 but an impossibly high Rating Count. These were filtered out using: data = data[data['Rating Count'] < data['Maximum Installs']]

Date Columns Converted to datetime

The 'Released', 'Last Updated' and 'Scraped Time' columns were converted from string to proper datetime objects, enabling date-aware SQL comparisons.

Maximum Installs Used Directly

No normalisation of install counts was needed. The 'Maximum Installs' numeric column was used as-is for all SQL aggregations and comparisons.

No Duplicate App Entries Found

There were no duplicate App Id records in the dataset. Each app appears exactly once.

Free / Paid — No Inconsistencies

The 'Free' boolean and 'Price' columns were fully consistent across all rows. CAST was used in SQL where arithmetic on these fields was required.

3. Exploratory Data Analysis

3.1 Install & Popularity Distribution

Total installs are highly skewed. The Tools category dominates with over 119 billion installs — roughly 65% more than the second-place Communication category. Beyond the top 5, the distribution forms a classic long tail with hundreds of niche categories each contributing a tiny fraction of total installs.

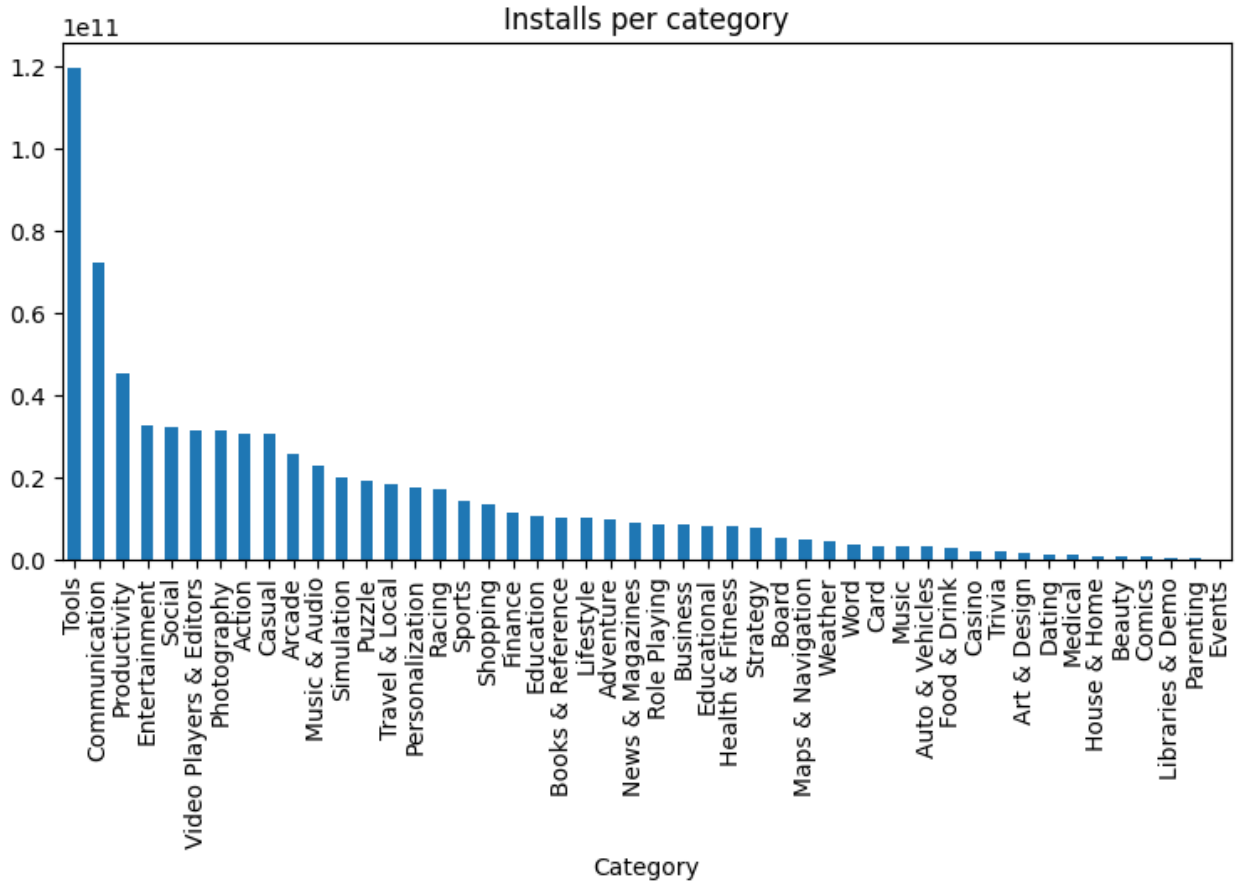

Figure 1 · Total installs per category (installs, summed)

Looking at raw app count per category tells a different story. Education and Music & Audio have the largest number of published apps, but their install volumes lag far behind Tools and Communication. This gap between app count and install volume highlights the uneven reach within the marketplace.

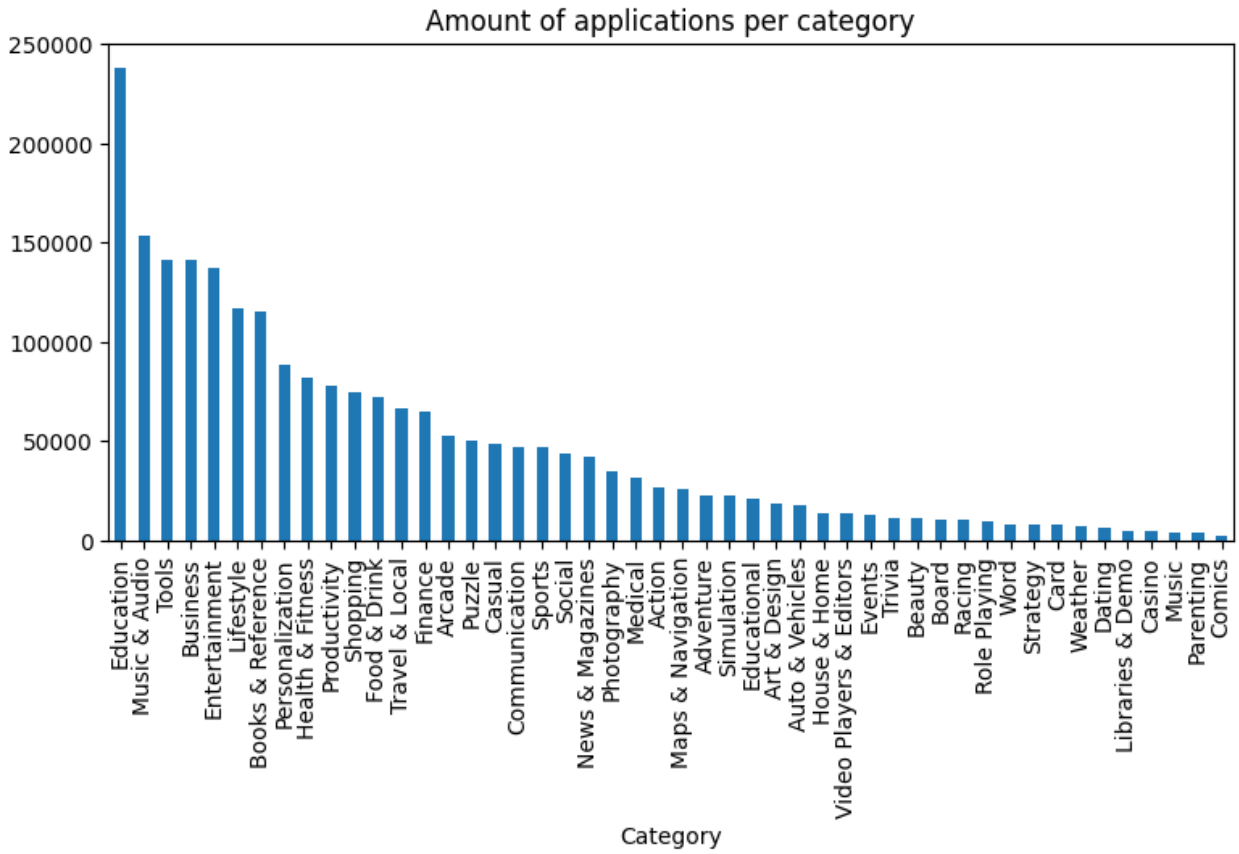

Figure 2 · Number of apps per category (sorted by app count)

3.2 Rating Analysis

Average ratings vary less across categories than one might expect — most sit between 3.8 and 4.4. The KDE of individual app ratings reveals a sharp spike at 5.0 which is suspicious: it is almost entirely driven by apps with very few reviews, where a single 5-star rating produces a perfect average. This is directly addressed in Q8.

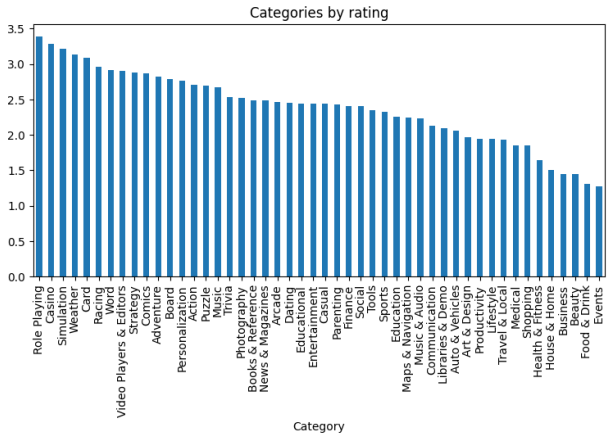

Figure 3 · Average rating per category

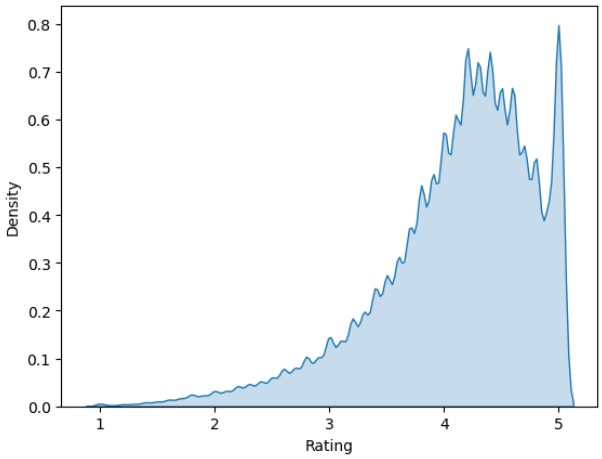

Figure 4 · KDE of app ratings (apps with rating > 0)

Filtering to only apps with fewer than 100 reviews makes the 5-star inflation even more visible. The distribution shifts heavily right, with a dramatic spike at 5.0. This confirms that the overall rating distribution is heavily skewed by low-engagement apps where one or two perfect ratings push the average to the maximum.

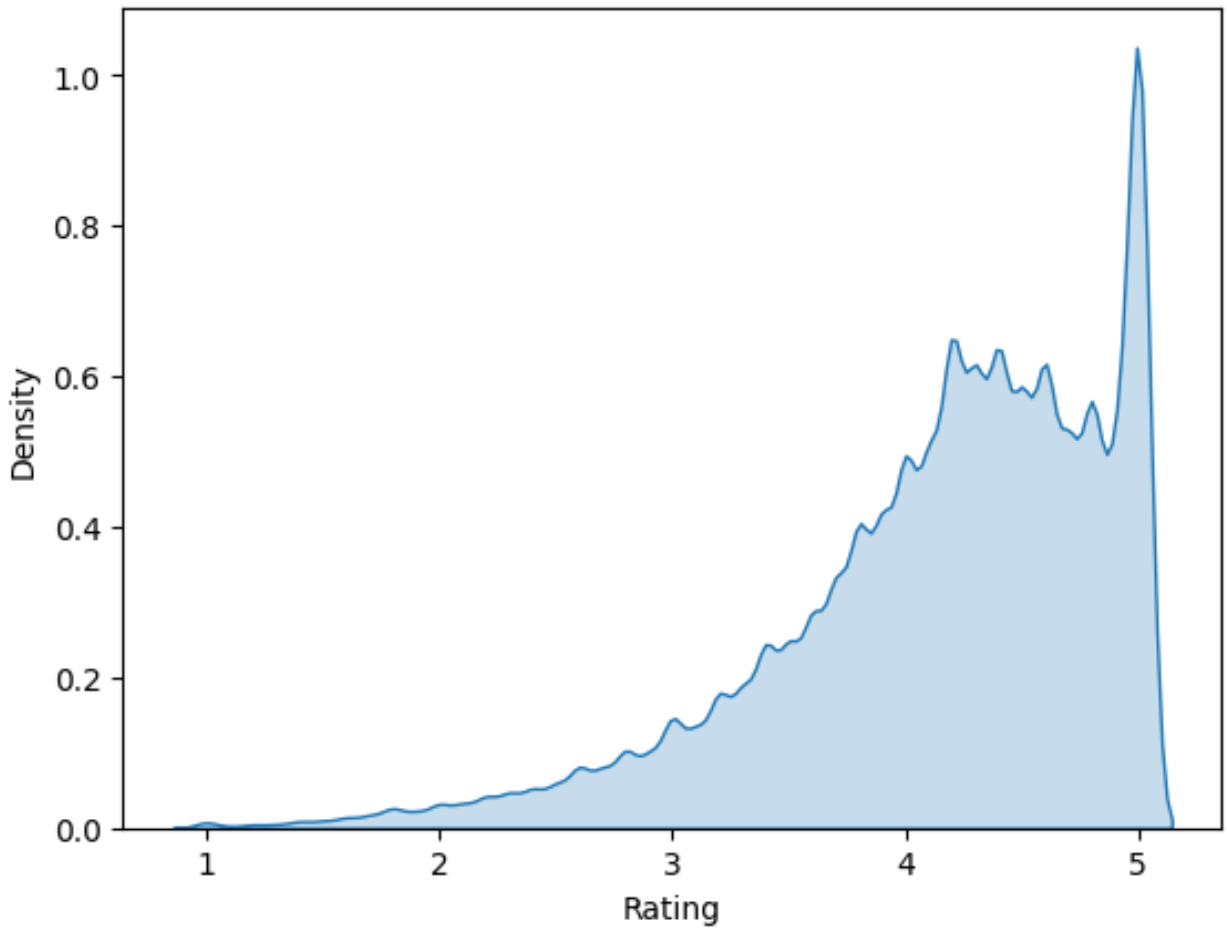

Figure 5 · KDE of ratings for apps with fewer than 100 reviews

3.3 Free vs Paid App Analysis

The Play Store is overwhelmingly free-to-download: fewer than 5% of apps in most categories are paid. Role Playing and Personalization categories have the highest paid-app proportions, but even there the ratio rarely exceeds 7%. This confirms the freemium model's dominance — monetisation happens primarily through ads and in-app purchases rather than upfront pricing.

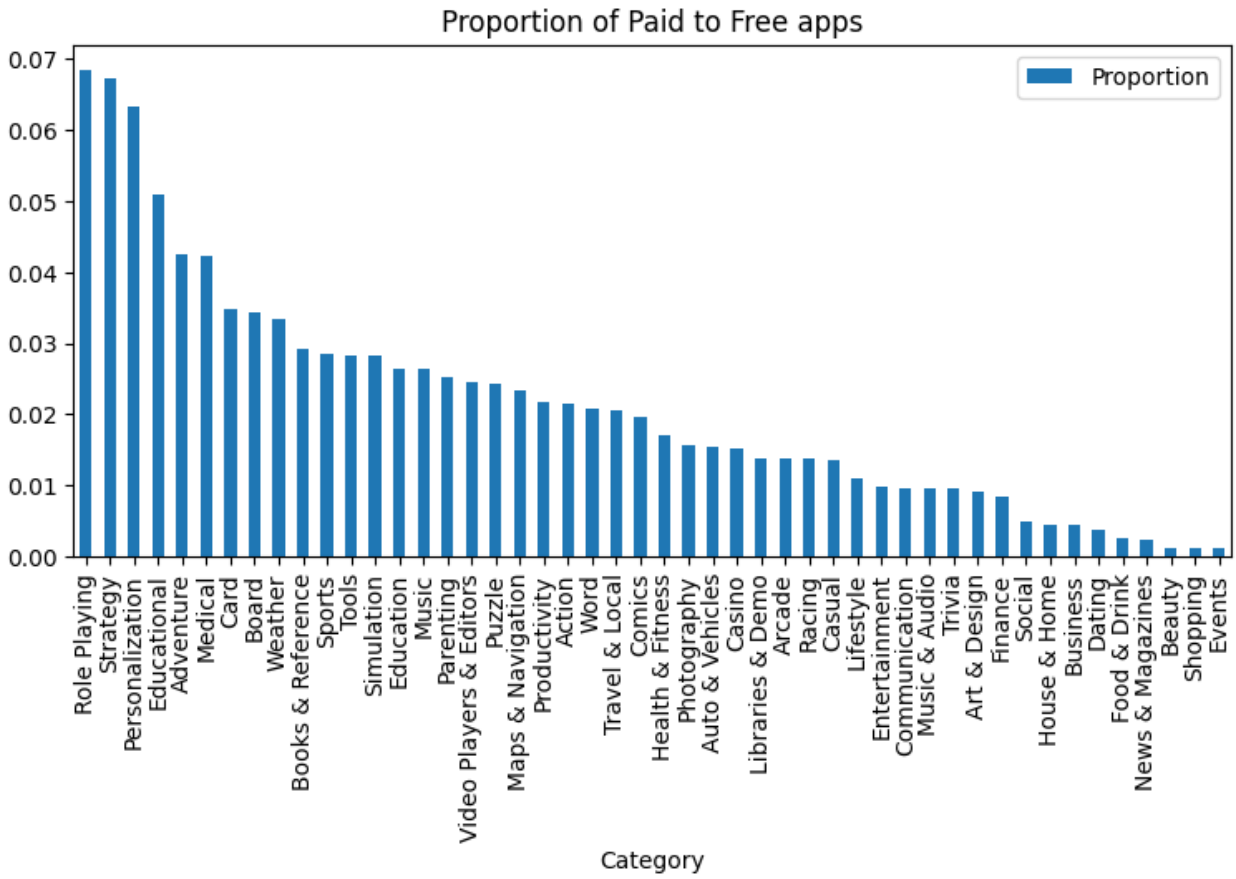

Figure 5 · Proportion of paid apps per category (top 30 by paid count)

When comparing average ratings between free and paid apps within each category, the differences are generally small. However, in some categories paid apps consistently outperform free equivalents in rating — possibly because users who pay self-select for higher-quality apps and then engage more carefully.

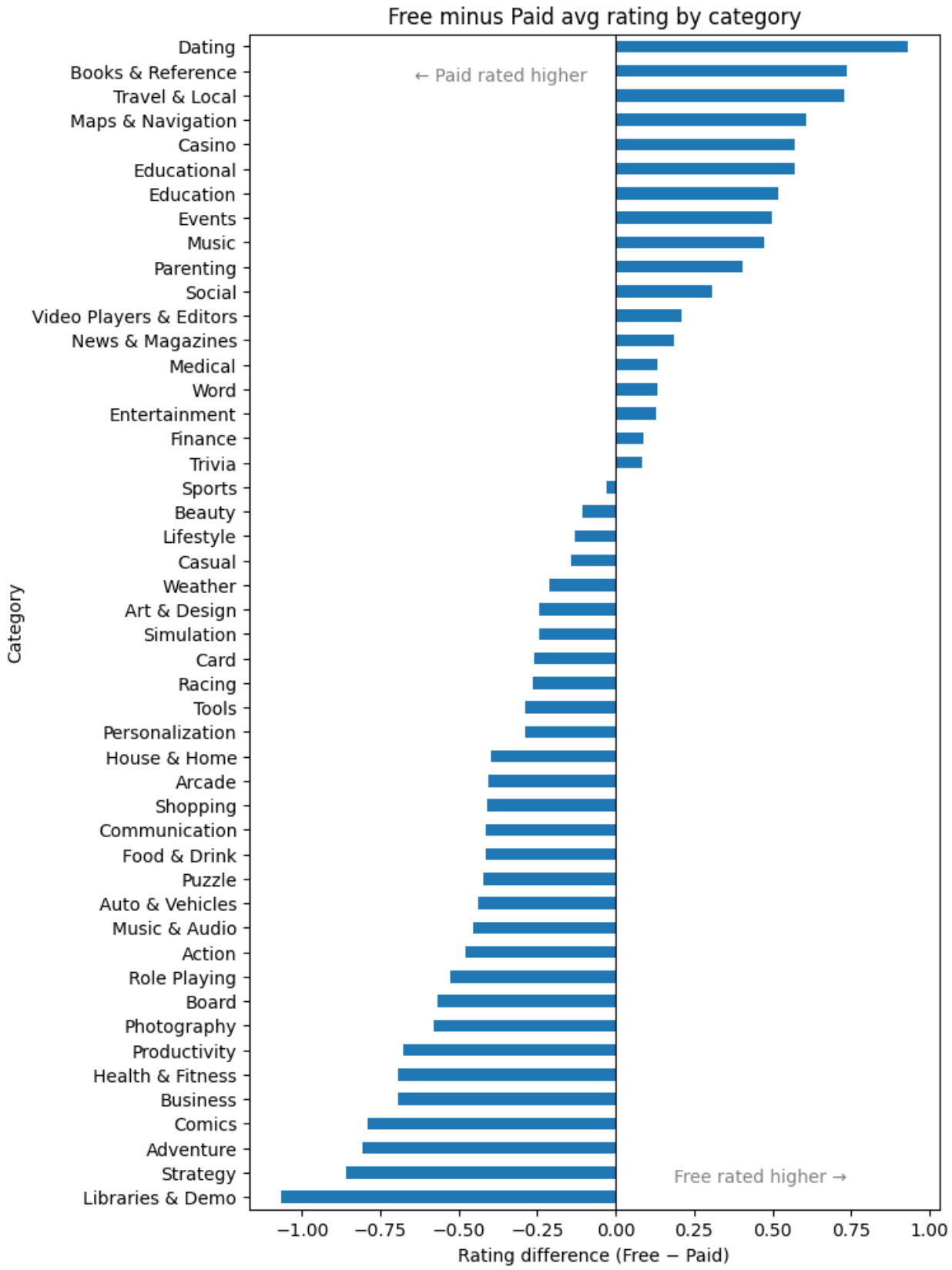

Figure 6 · Rating difference (paid minus free) per category

Plotting each category as a point — with Free rating on the x-axis and Paid rating on the y-axis — makes the relationship even clearer. Most categories cluster near the diagonal, meaning free and paid apps

perform almost identically. The few outliers above the line (e.g. Role Playing, Strategy) confirm that certain niches genuinely reward paid apps with higher satisfaction.

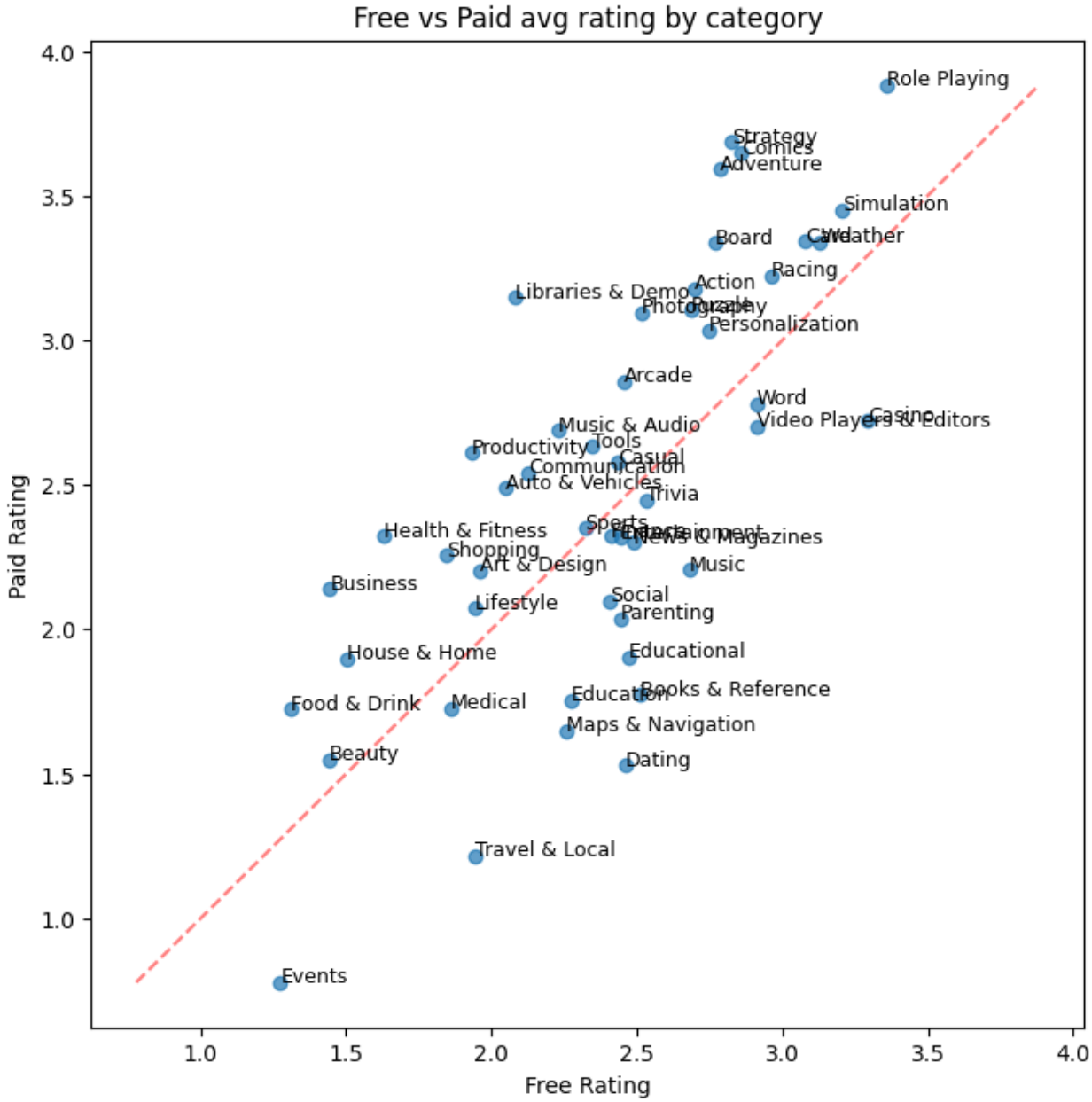

Figure 7 · Free vs Paid avg rating by category (points above diagonal = paid rates higher)

3.4 Developer Analysis

The relationship between a developer's total app output and their average app rating follows an inverted pattern past a certain threshold. Developers with fewer than ~15 apps show a wide spread of ratings. Once a developer publishes more than ~20–30 apps, their average rating tends to **decrease**. This suggests that mass publishing hurts quality — developers who focus on fewer, more carefully crafted apps achieve better user satisfaction scores.

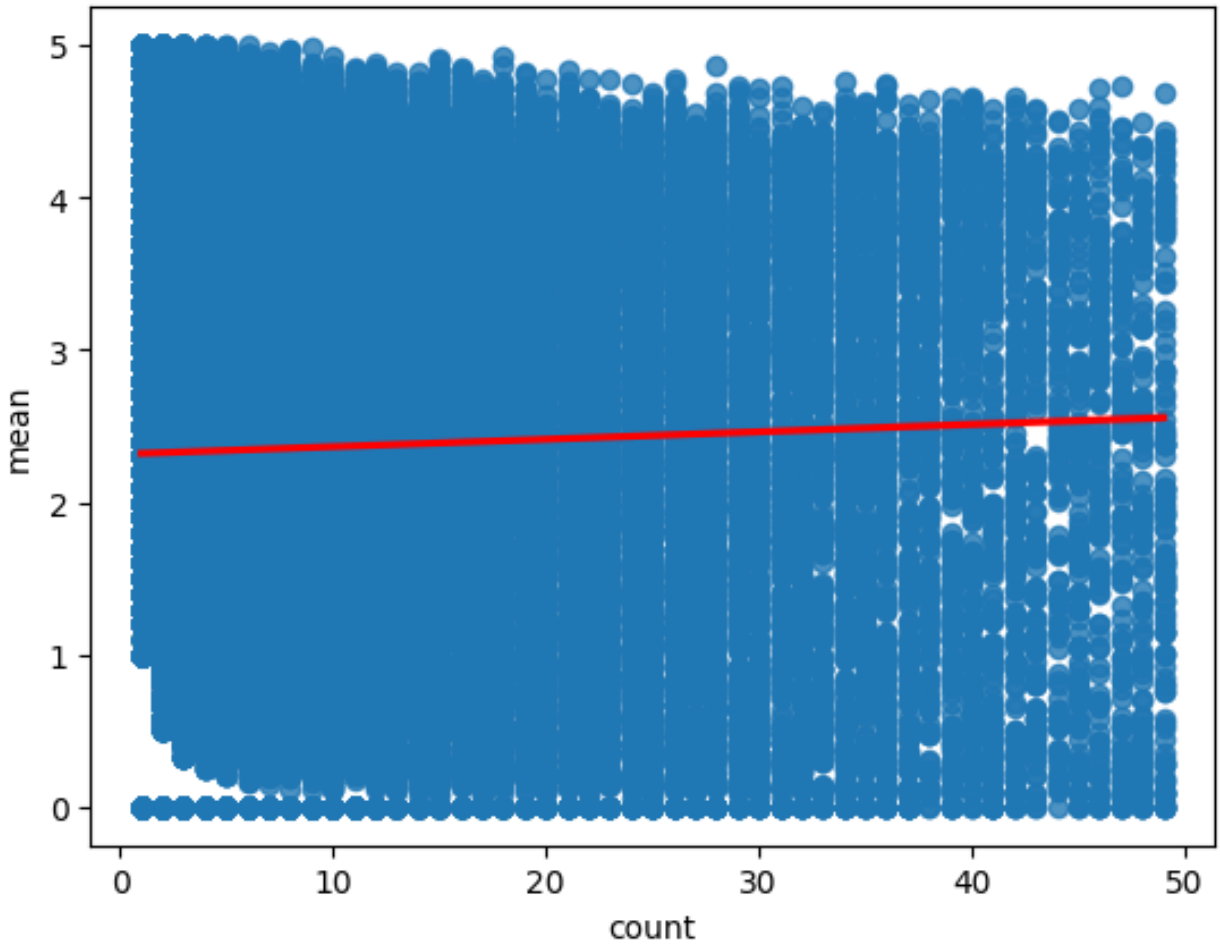

Figure 9 · Developer average rating vs. number of apps published (developers with fewer than 50 apps; regression line shown)

Removing extreme outliers and extending the view to developers with up to 500 apps makes the downward trend unmistakable. The regression line shows a clear negative slope: more apps published correlates with lower average rating, reinforcing that volume-focused publishing strategies come at a measurable quality cost.

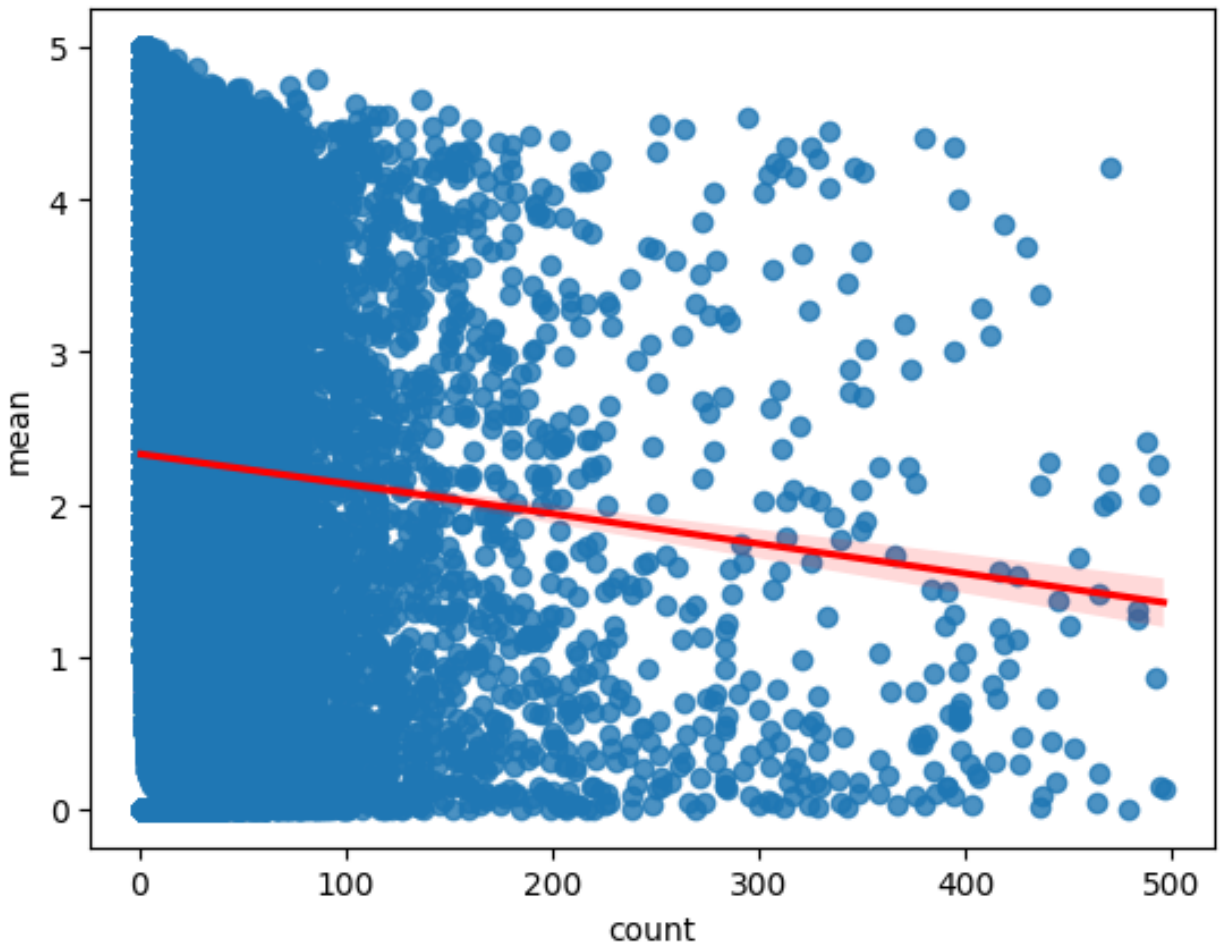

Figure 10 · Developer app count vs avg rating — outliers removed (count < 500)

3.5 Price Distribution & Content Ratings

For paid apps, price is heavily concentrated at psychological price points. The KDE shows a dramatic spike just below $1, confirming $0.99 as the dominant price point by a wide margin. Prices above $10 exist but represent a negligible share of the market.

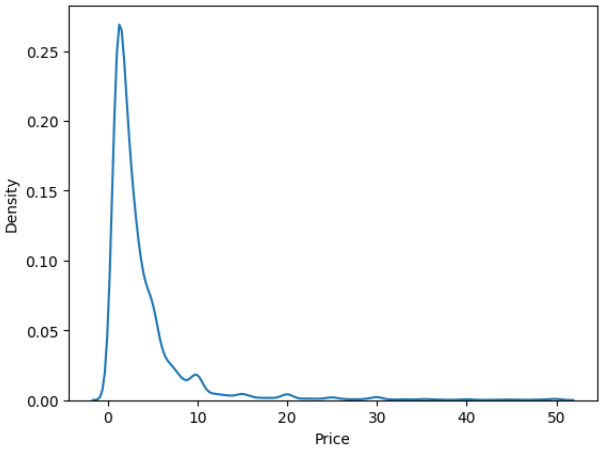

Figure 8 · KDE of paid app prices (filtered to price < $50)

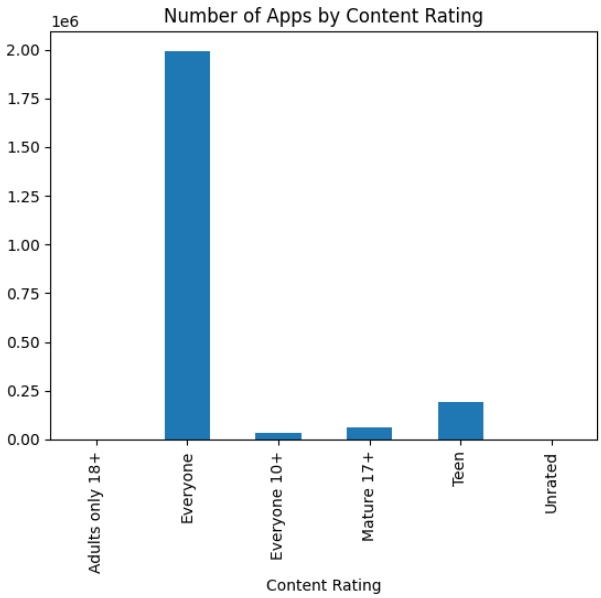

Figure 9 · Number of apps by content rating

The content rating distribution confirms the Play Store's family-friendly orientation: nearly 2 million apps are rated 'Everyone', dwarfing the Teen (~193K) and Mature 17+ (~60K) categories. This has direct implications for app design — the vast majority of developers target the broadest possible audience.

4. SQL Business Questions

The following ten SQL queries were written to answer concrete business questions about the Google Play Store dataset. Each query is presented with its full SQL code, the actual output from the database, and a brief analytical interpretation.

Q1

What are the Top 10 categories by total installs?

SQL Query

```sql
-- Top 10 categories by total installs
SELECT
category,
SUM(maximum_installs) AS total_installs
FROM google_play_store
GROUP BY category
ORDER BY total_installs DESC
LIMIT 10;
```

Query Output

Category	Total Installs
Tools	119.8 B
Communication	72.3 B
Productivity	45.4 B
Entertainment	32.5 B
Social	32.2 B
Video Players & Editors	31.4 B
Photography	31.2 B
Action	30.7 B

Insight

Tools leads by a wide margin with 119.8 B total installs, followed by Communication and Productivity. The long tail of niche categories shows how fragmented the market is beyond the top 3.

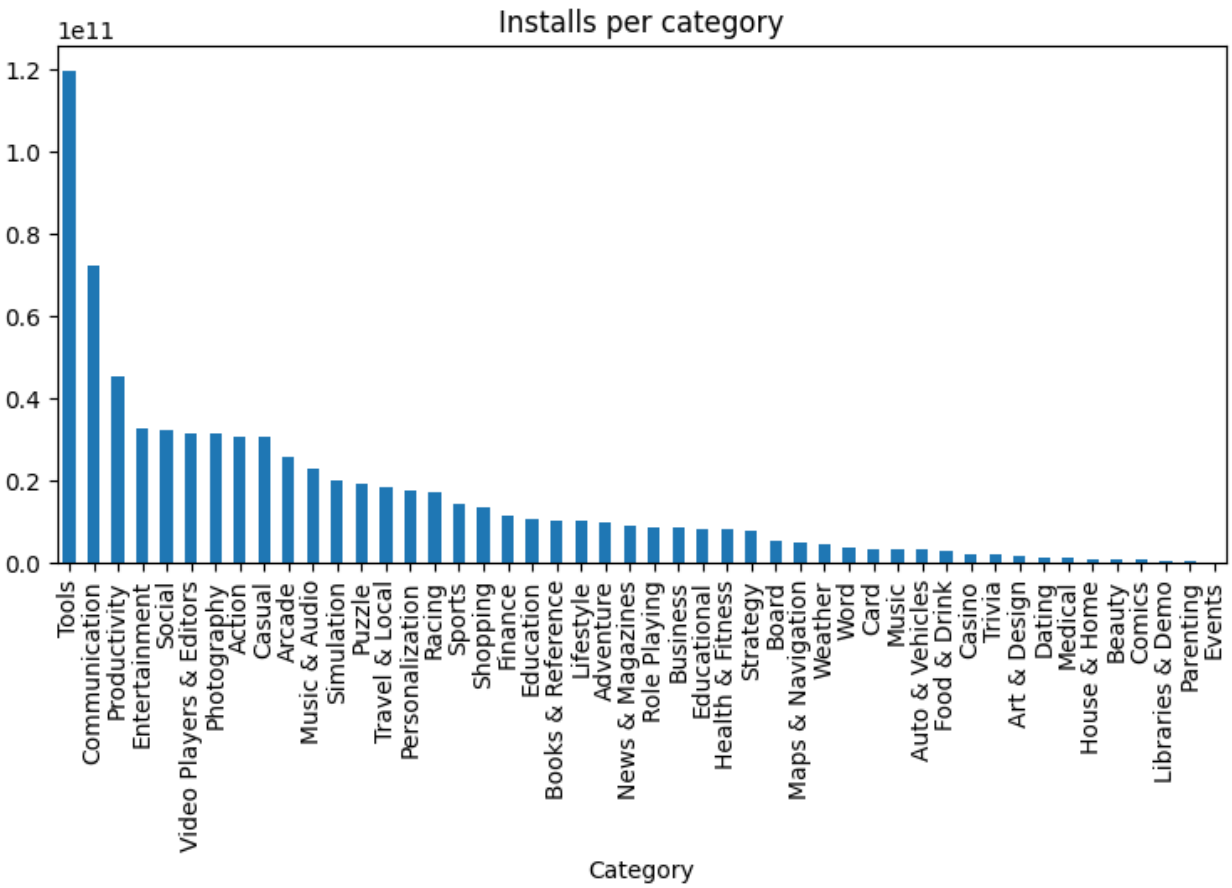

Figure 10 · Total installs per category — confirming Tools as the dominant segment

Q2

Which app is the best-rated per category (within the Top 10 most installed)?

SQL Query

```sql
-- Create a view: best app per category by rating
CREATE VIEW best_app_per_category AS
SELECT app_id, app_name, category, rating,
RANK() OVER (
PARTITION BY category
ORDER BY rating DESC, rating_count DESC
) AS rnk
FROM google_play_store;

-- Join with Top 10 install categories
SELECT cat_totals.category,
cat_totals.total_installs,
b.app_name,
b.rating
FROM (
SELECT category,
SUM(maximum_installs) AS total_installs
FROM google_play_store
GROUP BY category
ORDER BY total_installs DESC
LIMIT 10
) AS cat_totals
JOIN best_app_per_category AS b
ON b.category = cat_totals.category
AND b.rnk = 1
ORDER BY cat_totals.total_installs DESC;
```

Query Output

Category	Total Installs	App Name	Rating
Tools	119.8 B	QR code scanner	5
Communication	72.3 B	Cooperations office	5
Productivity	45.4 B	Calculator Plus	5
Entertainment	32.5 B	Maps Master for Minecraft	5
Social	32.2 B	DJJS	5
Video Players & Editors	31.4 B	Glitch Video Effect - Glitch FX	5
Photography	31.2 B	Photo Frame	5
Action	30.7 B	Hang Climb Adventure-Grand Mountain Climber	5

Insight

RANK() OVER (PARTITION BY category) finds the best-rated app inside each category. The result is then joined with the Top 10 categories by installs to see which app tops each major segment. Perfect ratings (5.0) appear frequently — reflecting apps with very few reviews rather than genuine universal acclaim.

Q3

Which categories have the highest proportion of paid apps?

SQL Query

```
-- Proportion of paid vs free apps by category
SELECT
category,
SUM(CASE WHEN free <> 'True'
THEN 1 ELSE 0 END) AS paid_amount,
COUNT(*) AS total_amount
FROM google_play_store
GROUP BY category
ORDER BY
CAST(paid_amount AS FLOAT)
/ total_amount DESC
LIMIT 10;
```

Query Output

Category	Paid Amount	Total Amount
Role Playing	660	9631
Strategy	553	8221
Personalization	5624	88,777
Educational	1070	20,996
Adventure	969	22,733
Medical	1330	31,544
Card	279	8033
Board	357	10,414

Insight

Role Playing leads with roughly 7% paid apps. Most categories stay well below 5%, confirming that free is the dominant model across the entire Play Store. Niche and premium-feeling categories (personalisation, simulation games) are the exception.

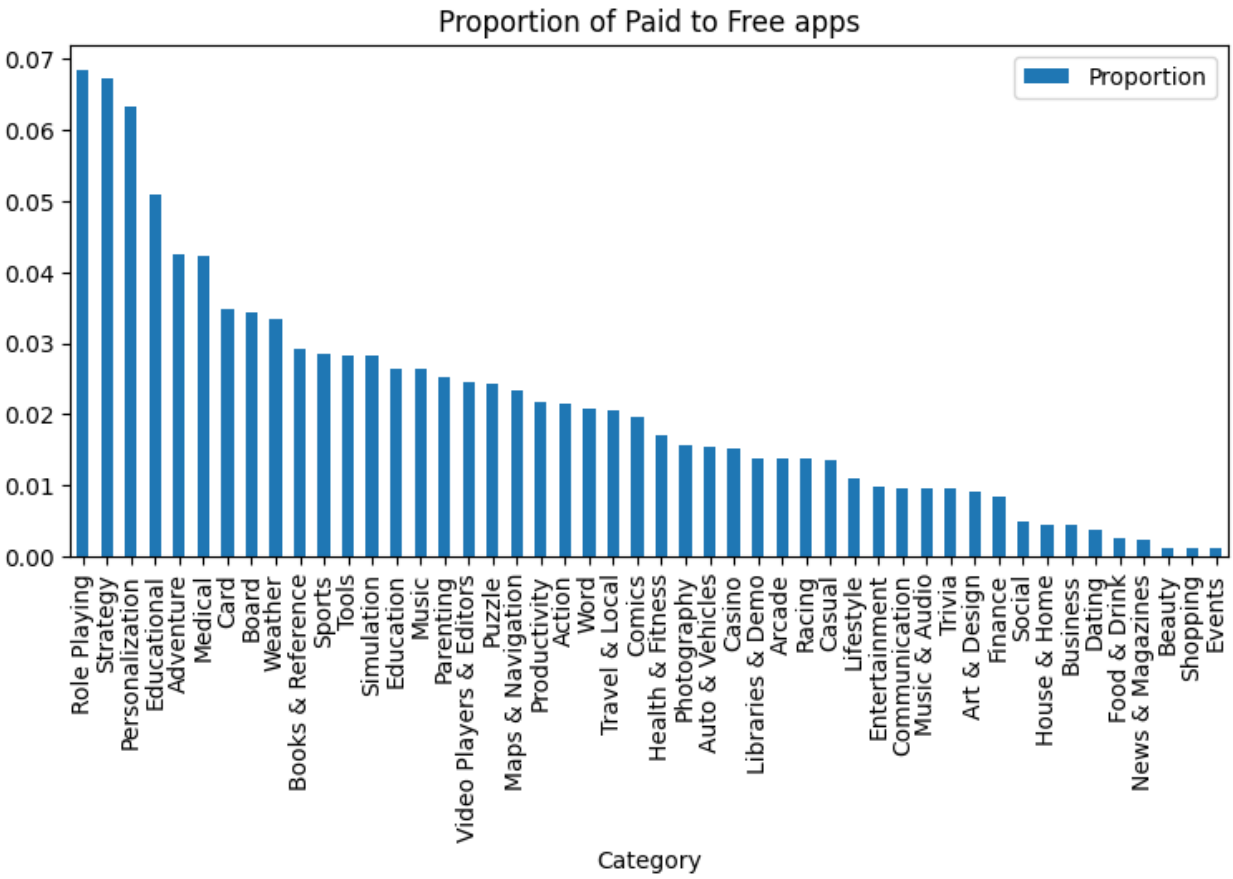

Figure 11 · Paid-to-free ratio per category from Python EDA

Q4

Which developers have published the most apps on the Play Store?

SQL Query

```
-- Top 10 developers by number of published apps
SELECT
developer_id,
COUNT(*) AS total_apps
FROM google_play_store
GROUP BY developer_id
ORDER BY total_apps DESC
LIMIT 10;
```

Query Output

Developer Id	Total Apps
Subsplash Inc	5394
TRAINERIZE	4923
ChowNow	4808
OrderYOYO	2831
Phorest	2743
BH App Development Ltd	2209
Sharefaith	2041
Flipdish	1938

Insight

Grouping by developer_id and counting rows uncovers prolific publishers. High app counts may signal content farms or SDK testing accounts — a useful quality filter for any recommendation system.

Q5

Which developers have the most total installs across all their apps?

SQL Query

```
-- Top 10 developers by total installs
SELECT
developer_id,
SUM(installs) AS total_installs
FROM google_play_store
GROUP BY developer_id
ORDER BY total_installs DESC
LIMIT 10;
```

Query Output

Developer Id	Total Installs
Subsplash Inc	625,253
ChowNow	335,424
OrderYOYO	290,641
Branded Apps by MINDBODY	210,893
Phorest	173,053
Flipdish	171,439
Apptegy	168,173
Magzter Inc.	154,083

Insight

The top developers by total installs are largely the same names as in Q4. The order shifts slightly — a few highly viral apps can push a developer up the ranking — but the dominant publishers remain almost identical in both queries.

Q6

Which apps are underrated — high review engagement but low installs?

SQL Query

```
-- Underrated apps:
-- high review ratio relative to installs
SELECT
app_name,
rating_count,
maximum_installs
FROM google_play_store
ORDER BY
CAST(rating_count AS FLOAT)
/ maximum_installs DESC
LIMIT 10;
```

Query Output

App Name	Rating Count	Maximum Installs
Hi Speed Radar,Camera, Blitzi, Traffic, Alert…	224	225
All in one Status Saver: Save HD Images, Videos	95	96
Choper Slice - Master Chef	81	82
LigaLigura: The Ultimate Precious Metal Calcu…	146	148
McPAID	50	51
Chase Craze	93	95
All Status Saver - Download videos & Images	39	40
DBOH Converter (Decimal to Binary, Octal, Hexal)	39	40

Insight

Ordering by the ratio of rating_count to maximum_installs surfaces apps where almost every installer also left a review. A ratio close to 100% is suspicious — reviews may have been written by friends or purchased on third-party sites rather than earned organically.

Q7

Which apps are overhyped — massive installs but low ratings?

SQL Query

```sql
-- Step 1: avg installs of top 5% of apps
WITH total AS (
SELECT
CAST(COUNT(*) * 0.05 AS INTEGER)
AS top_n
FROM google_play_store
)
SELECT AVG(maximum_installs) AS avg_installs
FROM google_play_store
ORDER BY maximum_installs DESC
LIMIT (SELECT top_n FROM total);
-- Result: ~324,704

-- Step 2: overhyped apps
SELECT *
FROM google_play_store
WHERE maximum_installs > 324704
AND rating_count > 0
ORDER BY rating ASC
LIMIT 10;
```

Query Output

App Name	Category	Rating	Maximum Installs
Offroad car driving:4x4 off-road rally legend…	Racing	1	331,076
TV9 Telugu	News & Magazines	1.10	2.2 M
Thingiverse	Social	1.10	659,787
Android TV Home	Tools	1.10	60.7 M
Destiny Teller	Entertainment	1.10	1.4 M
Face Symmetry	Tools	1.10	579,733
BOTOLA	Sports	1.10	1.1 M
Fm Transmitter Car 2.1	Auto & Vehicles	1.20	478,657

Insight

A CTE first computes the average installs for the top 5% of apps (~324K). Apps above this threshold with at least one review are then sorted by rating ascending. These apps have viral reach but consistently fail to satisfy their users.

Q8

What percentage of apps have fewer than 100 reviews?

SQL Query

```
-- % of apps with fewer than 100 reviews
WITH stat AS (
SELECT COUNT(*) AS few_reviews
FROM google_play_store
WHERE rating_count < 100
)
SELECT
few_reviews,
(SELECT COUNT(*) FROM google_play_store)
AS total_count,
ROUND(
(CAST(few_reviews AS FLOAT)
/ (SELECT COUNT(*)
FROM google_play_store))
* 100, 2
) AS percentage
FROM stat;
```

Query Output

Few Reviews	Total Count	Percentage
1.9 M	2.3 M	82.40

Insight

82.4% of apps have fewer than 100 reviews — meaning the vast majority go largely unnoticed. Discoverability is one of the biggest challenges on the Play Store. Only 17.5% of apps accumulate meaningful review volume.

Q9

What are the most common price points for paid apps? Is there a 'sweet spot'?

SQL Query

```
-- Most common price points for paid apps
SELECT
price,
COUNT(*) AS app_count
FROM google_play_store
WHERE price > 0
GROUP BY price
ORDER BY COUNT(*) DESC
LIMIT 10;
```

Query Output

Price	App Count
0.99	11,455
1.99	5687
2.99	3818
1.49	3683
4.99	2434
3.99	2357
2.49	2140
3.49	1246

Insight

$0.99 is the clear mode with 11,455 apps — more than twice the second price point. The distribution confirms a strong 'sweet spot' at $0.99, with $1.99 and $2.99 as secondary peaks. Prices above $5 are rare. The KDE visualisation reinforces this with a dramatic spike just below $1.

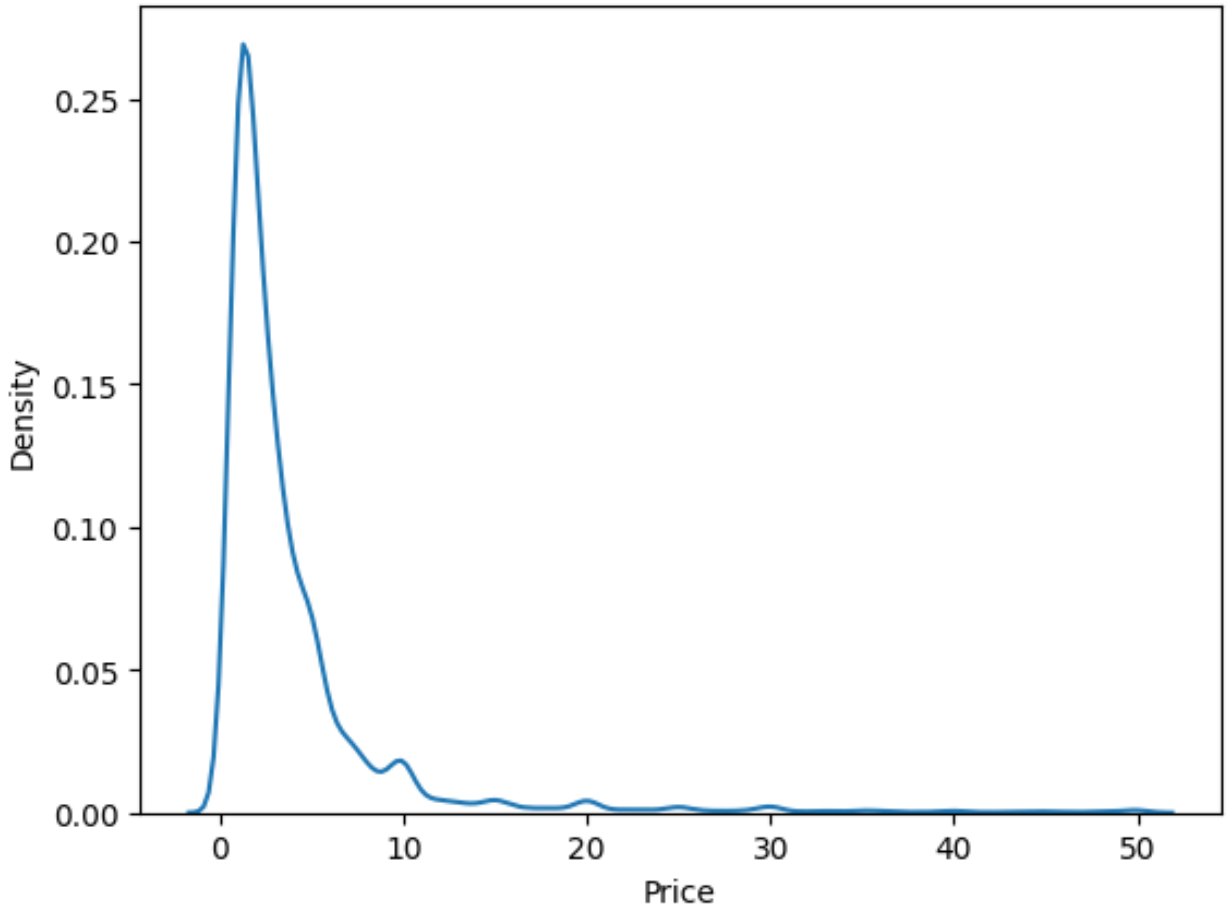

Figure 12 · KDE of paid app prices (price < $50) — spike at $0.99 clearly visible

Q10

Which content rating (Everyone, Teen, Mature 17+) dominates by number of apps?

SQL Query

```sql
-- Content rating distribution
SELECT
content_rating,
COUNT(*) AS app_count
FROM google_play_store
GROUP BY content_rating
ORDER BY COUNT(*) DESC;
```

Query Output

Content Rating	App Count
Everyone	2.0 M
Teen	192,750
Mature 17+	59,608
Everyone 10+	33,085
Unrated	154
Adults only 18+	132

Insight

'Everyone' has ~2.0M apps — dwarfing all other ratings. Teen is a distant second with ~193K. This confirms the Play Store is overwhelmingly family-friendly by design, reflecting developers' strategy to maximise their accessible audience.

5. Key Conclusions

- **Tools, Communication & Productivity dominate**
  These three categories account for the largest share of total installs on the Play Store — with Tools alone exceeding 119 billion installs. High install volume does not correlate with high app count: Tools has far fewer apps than Education.

- **Most apps have very little engagement**
  82.4% of apps have fewer than 100 reviews, showing that discoverability is a major challenge. The platform's long tail means that most apps are functionally invisible to the average user.

- **Paid apps cluster at psychological price points**
  $0.99, $1.99 and $2.99 dominate the paid app market, confirming that low prices reduce friction for paid downloads. Very few apps price above $10 and survive with significant install volume.

- **Prolific developers = Most-installed developers**
  Q4 and Q5 surface almost identical top names — high app output and high install reach go hand in hand. However, Python EDA shows that past a certain threshold, publishing more apps correlates negatively with average rating: volume eventually hurts quality.

- **'Everyone' content rating dominates**
  Over 87% of apps target all age groups. Developers overwhelmingly choose the broadest possible audience classification, reflecting a market-reach-first strategy over niche targeting.

This report was produced as part of the IronHack Data Science & Machine Learning Bootcamp Business Challenge. All SQL queries were executed against a local SQLite database derived from the Google Play Store Kaggle dataset. Python visualisations were generated using pandas, matplotlib and seaborn.